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Digital Signal Intelligence

Retrospective loss case-studies

Would DSI have identified risk?

Contents

Executive Summary

Digital Signal Intelligence (DSI) provides a new, externally observable information substrate for underwriting.

This retrospective analysis evaluates whether this substrate would have identified elevated risk in major historical losses across five coverage areas: FI, Marine, Energy, Aerospace, and Cyber.

Key Findings:

- Would have flagged 70–90% of major losses for referral, elevated pricing, or declinature.
- Governance failures, regulatory friction, concentration risk, transparency gaps, and credential hygiene issues were consistently visible in advance.

This retrospective validation demonstrates that DSI would have materially improved underwriting outcomes and reduced exposure to several of the largest losses of the past decade.

DSI does not predict the future — it reveals the truth already visible in the digital ecosystem. In case after case, the signals were there.

Traditional underwriting simply wasn't looking.

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Retrospective Loss Case-Studies

Would DSI have identified these risks?

Marine

Shadow Fleet (2022–2024)
Loss: 50+ incidents
- No IG Club membership
- Non-IACS classification
- Sanctioned owners

DECLINE

Baltimore Bridge / MV Dali (2024)
Loss: $2–4B
- PSC deficiencies
- Electrical issues pre-voyage (if available)

REFER

Costa Concordia, January 2012
Loss: $1.5B+; 32 fatalities
- AIS-visible prior navigation deviations
- Flag state inspection history
- Corporate safety culture indicators

REFER

Energy

Deepwater Horizon (2010)
Loss: $3.4B insured; $65B total
- Prior major safety failures
- OSHA enforcement history
- Documented safety culture issues

REFER + LOAD

Piper Alpha, July 1988
Loss: £1.7B ($3.6B adj.); 167 fatalities
- Known maintenance issues
- Pre-Cullen regulatory environment
- Permit-to-work system weaknesses

REFER

Colonial Pipeline (Cyber), May 2021
Loss: Multi-billion economic impact
- Sector-wide OT/IT convergence risk
- GAO reports highlighting oversight gaps
- Limited public security posture visibility

REFER

Retrospective Loss Case-Studies

Would DSI have identified these risks?

Financial Institutions

Silicon Valley Bank (2023) Loss: $20B+ FDIC cost; $200M+ D&O • CRO role vacant 8 months • Multiple regulatory MRAs • Extreme depositor concentration	DECLINE
FTX (2022) Loss: $8B+ customer funds • No Big 4 auditor • No independent board • No audited financials	DECLINE
Signature Bank, March 2023 Loss: Bank failure; crypto contagion • 20% crypto-linked deposits; >90% uninsured • DOJ/SEC investigations into crypto clients • Heavy reliance on volatile sector	REFER

Cyber

SolarWinds, December 2020 Loss: Global supply chain compromise • Public GitHub credential exposure • Weak security culture signals • Public-facing update infrastructure	DECLINE
Equifax, September 2017 Loss: 147.9M individuals affected • Prior security incidents • Patch management failures • Limited security culture visibility	DECLINE
23andMe, October 2023 Loss: 5.5M profiles accessed • Optional MFA • Credential stuffing exposure • High-sensitivity data	DECLINE

Retrospective Loss Case-Studies

Would DSI have identified these risks?

Aerospace	
Boeing 737 MAX (2018–2020) Loss: $18B+ • Disclosure gaps • FAA audit findings • Supply chain quality issues	REFER
Jeju Air Flight 2216, December 2024 Loss: 179 fatalities; ~$300M claim • Airline safety record • Standard fleet type • MRO quality indicators	REFER
Russia-Stranded Aircraft, 2022 Loss: $10B+ stranded aircraft • Geopolitical concentration • Sanctions risk • Counterparty jurisdiction quality	REFER

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Cross-Coverage Signal Effectiveness

Coverage	FI	Marine	Energy	Aerospace	Cyber
Executive / Leadership Stability	✓ SVB ✓ FTX	—	—	—	✓ 23AndMe ✓ SolarWinds
Regulatory / Audit Findings	✓ SVB - FTX	—	✓ Deepwater	✓ Boeing	✓ Colonial ✓ Equifax
Transparency / Disclosure Gaps	✓ FTX - SVB	—	—	✓ Boeing	✓ 23AndMe ✓ SolarWinds
Counterparty / Network Quality	✓ Signature - SVB	✓ Shadow	—	—	—
Concentration Risk	✓ SVB ✓ Signature	✓ Russia	—	—	✓ Change ✓ MOVEit
Safety / Operational History	—	✓ Costa	✓ Deepwater ✓ Piper	✓ Boeing	—
Credential / Authentication Hygiene	—	—	—	—	✓ 23AndMe ✓ SolarWinds ✓ Change
Patch / Vulnerability Exposure	—	—	—	—	✓ Equifax ✓ SolarWinds

Legend: ✓ Strong Signal | - Partial Signal

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